Assignment A: Make Customer Support Scale for Non-Technical Users

Cheerio Copilot: Autonomous Diagnostics, 1-Click AutoFix & Warm Escalation

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Submission Links: 📊 **Google Sheet (Formulas Open):** [Clickable Google Sheet Link] 🚀 **Clickable Prototype:** [Interactive Prototype Demo]
📽️ **Loom Walkthrough (3:45):** [Loom Video Link]

Context & Executive Summary: Cheerio AI empowers 5,000+ businesses across WhatsApp, SMS, and Email. However, non-technical Indian SMB merchants (D2C founders, coaching institutes) struggle with Meta Cloud API verification, template rejections, and chatbot logic errors—generating ~9,000 monthly tickets. We propose **Cheerio Copilot**: an in-workflow AI diagnostics engine that inspects Meta Graph API telemetry, translates errors into plain English/Hinglish, and resolves 66% of queries via **1-Click AutoFix**, saving **₹48.1 Lakhs/year (-74%)**.

PART 1: SOLUTION STRATEGY — USERS, QUERIES & PRIORITIZATION

1. Target User Archetypes & Top 5 Support Bottlenecks

Users fall into two profiles: (1) **D2C Founder ("Ramesh")** running marketing on mobile/laptop web, seeking quick festive sales; (2) **SMB Growth Marketer ("Sneha")** managing leads without any JSON or webhook knowledge.

Top 5 Issue / Query	Root Cause & Why Non-Technical Users Get Stuck	Business Impact & Priority
1. WhatsApp API & Embedded Signup	Number already active on WhatsApp app; unverified Business Manager; missing WABA payment card; forgotten 2-Step PIN.	**Highest Churn** 42% onboarding drop-off. If blocked for 48h, user requests refund.
2. Meta Template Rejection & Category	Missing sample values for `{{1}}`; variable at sentence start; promotional discounts misclassified as "Utility".	**Highest Volume** 32% of all tickets. Users blame Cheerio when Meta algorithms reject text.
3. Chatbot Flow Loops & Dead-Ends	Strict string condition (`reply == 'Yes'`) lacks fallback intent. Fails when customers type "yes please" or Hinglish.	**Revenue Loss** WhatsApp bot goes silent on high-intent inbound leads.
4. E-Commerce / CRM Webhooks	Expired Shopify access tokens; missing country codes (`+91`); unhandled abandoned checkout payload changes.	**Anxiety Trigger** High panic when automated abandoned cart recovery stops firing.
5. Broadcast Failures & Tier Limits	24-hr customer service window expired; depleted Meta wallet; reaching Meta Tier 1 daily limit (250 conversations/24h).	**Emergency Influx** Festive flash sale broadcasts fail, flooding live support chat.

2. Solution Alternatives & RICE Prioritization Matrix

We evaluated 4 strategic options using the **RICE Framework** (Reach, Impact, Confidence, Effort) to determine the highest-leverage solution:

Solution Alternative	Reach (1-10)	Impact (0.5-3)	Confidence	Effort (wks)	RICE Score	Strategic PM Decision
Option 1: Static Video Academy / Help Docs	7	1.0 (Low)	80%	4	**1.40**	Rejected: Non-technical SMBs do not read documentation when anxious.
Option 2: Rule-Based FAQ Decision-Tree Bot	8	1.5 (Med)	70%	5	**1.68**	Rejected: Extremely brittle; fails on typos, escalates 70%+ of queries.
Option 3: 1-on-1 Human Concierge Calls	3	3.0 (High)	90%	20	**0.41**	Rejected: Unit economics break; hiring 50 reps is unviable on ₹1,999 plans.
Option 4: Cheerio Copilot & Self-Healing Engine	**9**	**3.0 (Massive)**	**85%**	**6**	**3.83**	**Selected: In-workflow telemetry + 1-Click AutoFix + Warm Escalation.**

PART 1 (CONTINUED): SOLUTION ARCHITECTURE, GUARDRAILS & NORTH STAR

3. Description of Chosen Solution: Cheerio Copilot & Self-Healing Diagnostics

Unlike passive chatbots that send docs links, Cheerio Copilot operates as an **in-workflow autonomous agent** directly embedded into the user's workspace:

- **What It DOES:**
  1. *Proactive Workspace Telemetry:* Listens to Meta Graph API responses (e.g. Error 132000, 131031) and displays a "Setup Health Scorecard" before users submit tickets.
  2. *Bilingual Plain-Language Translation:* Translates technical errors into conversational English & Hinglish (e.g., *"Meta ne sample value na hone ki wajah se reject kiya"*).
  3. *1-Click AutoFix Actions:* Injects sample variables, reformats text prefixes, re-tags categories, and links fuzzy AI fallback nodes with a single user click.
  4. *Pre-Packaged Diagnostic Dossier:* Bundles WABA ID, phone ID, Meta error codes, and previous attempts into an automated ticket when escalating to humans.
- **What It Will NOT Do (Strict Guardrails & Boundaries):**
  1. *No Destructive Data Mutations:* Cannot delete customer contact lists, campaigns, or payment methods without 2FA confirmation.
  2. *No Overriding Meta Anti-Spam Bans:* Cannot magically unban numbers flagged by Meta; prepares an official Meta Appeal Packet with legal document verification instead.
  3. *No Financial Discretion:* Will not issue subscription refunds or alter plan prices; billing queries are escalated to human finance leads.
  4. *No Unconstrained Open Chat:* Rejects off-topic queries to prevent hallucination and maintain token frugality.

4. When Does a Human Still Need to Step In? (Warm Escalation Protocol)

Human agents step in only for high-friction, regulated, or emotionally sensitive scenarios:

Escalation Category	Trigger Event	AI Copilot Action Before Handoff	Human Agent Workflow (Zero Back-and-Forth)
Meta Account Suspension	WABA status: `RESTRICTED` / `BANNED`	Compiles business registration docs, WABA ID & logs.	Meta Partner Liaison submits priority appeal to Meta Partner Desk.
Billing & Commercial	Refund requests, invoice disputes, tier upgrades	Verifies subscription plan & payment gateway log.	Finance Lead reviews charge history and processes adjustment.
Sentiment & Anger Spikes	Sentiment score < 0.3 ("fraud", "talk to human")	Freezes AI immediately; acknowledges user urgency.	Senior Support Lead joins live chat within 3 minutes with full context.
Multi-Attempt Failure	2 failed auto-fixes in a single user session	Packages system state and screen recording.	L2 Technical Specialist diagnoses custom webhook or payload mismatch.

Impact: Because human agents receive a complete **Diagnostic Dossier**, Average Handling Time (AHT) drops from **18 minutes to 7 minutes (-61%)**.

5. Success Metrics & 90-Day North Star Target

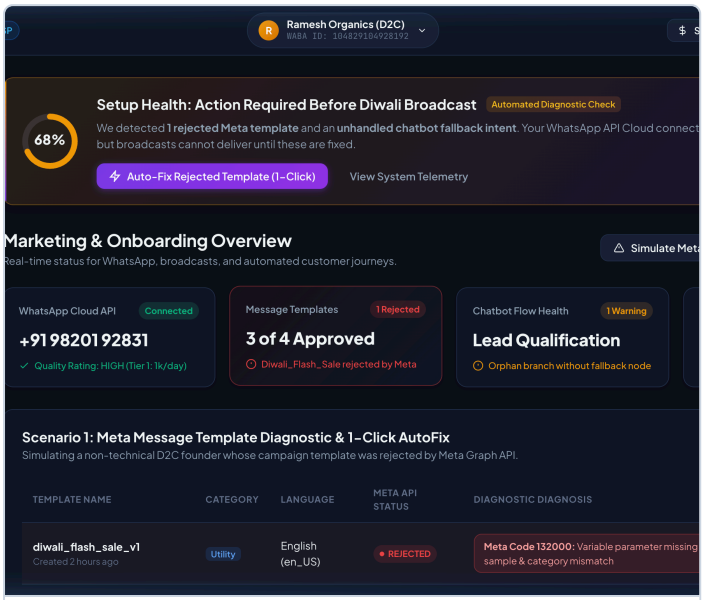
Metric Name	Baseline	Month 3 Target	Measurement Methodology & Value Driver
1. Autonomous Deflection Rate	0% (All human)	**65% - 70%**	Tickets resolved end-to-end by Copilot without reopening within 48 hours.
2. Time-to-First-Message (TTFM)	48 Hours	**< 2 Hours**	Elapsed time from signup to sending first verified WhatsApp campaign.
3. First Contact Resolution (FCR)	41%	**> 82%**	Combined AI instant resolution + 1-touch warm human escalation.
NORTH STAR: 7-Day Setup & Activation Rate	**35%**	**60% (+71% lift)**	**% of new businesses verifying WABA, getting 1 template approved, and activating 1 flow within 7 days.**

6. Execution Plan to Improve North Star Metric from 35% to 60%

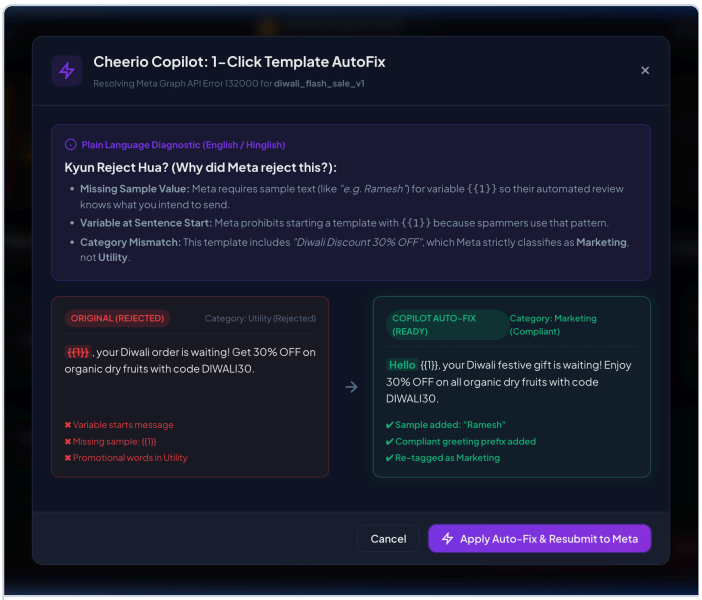
- **Month 1 (Telemetry & Template Engine):** Instrument Meta Graph API error codes (132000, 131031). Deploy Top 10 Template Auto-Fix rules (sample parameters, category re-tagging). *Expected Lift: +12%.*
- **Month 2 (Proactive HealthCheck & Hinglish NLP):** Launch the persistent "Setup Health Score" widget. Add bilingual Hinglish natural language parsing. Deploy Smart Escalation Dossiers. *Expected Lift: +8%.*
- **Month 3 (Integrations & Self-Healing Webhooks):** Launch 1-click Shopify/WooCommerce re-authorization and webhook diagnostic test pings. Implement prompt caching for sub-second latency. *Expected Lift: +5% (Achieving 60%).*

PART 2: CLICKABLE DESIGN PROTOTYPE (END-TO-END WALKTHROUGH)

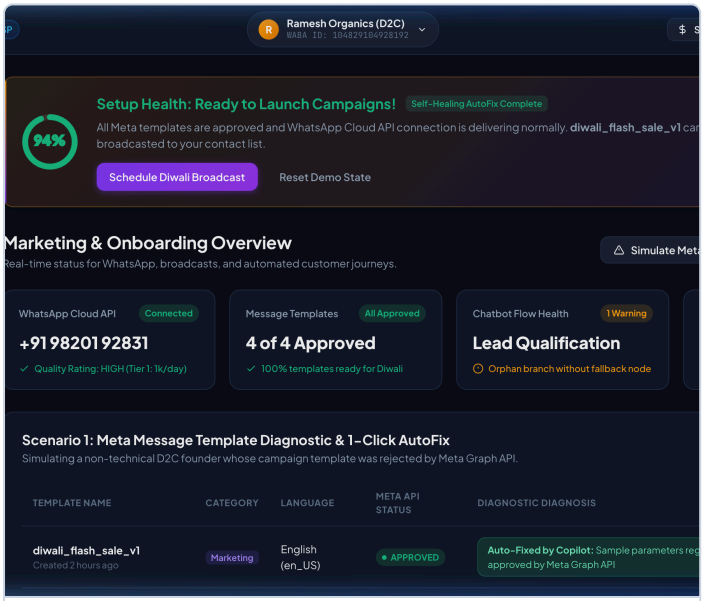
A fully functioning interactive prototype was built and tested at `http://localhost:3000`. It models the primary scenario of a non-technical D2C founder ("Ramesh") who encounters a Meta template rejection right before launching a Diwali flash sale.



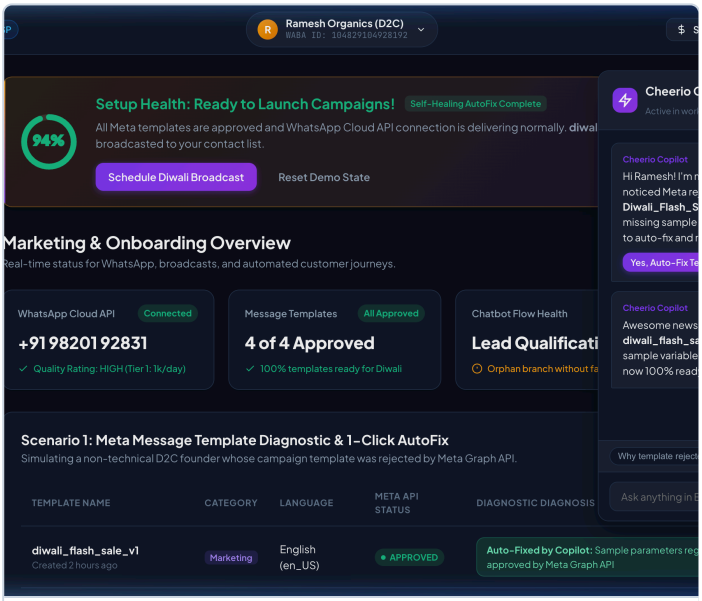
Screen 1: Proactive Setup Health Banner (68% Health, 1 Rejected Meta Template Detected)



Screen 2: Plain Language Diagnosis & Side-by-Side Diff (Original vs 1-Click AutoFix Preview)



Screen 3: Instant AutoFix Resolution (Meta Status: APPROVED, Health Score reaches 94%)



Screen 4: In-Context Copilot Chat Drawer (Bilingual English/Hinglish & Warm Escalation)

Step-by-Step User Journey Demonstrated in Prototype:

1. **Step 1 — Proactive Health Alert:** The user logs in and immediately sees a top alert: *"Setup Health: 68% — 1 Rejected Meta Template"*. Instead of waiting for failure, Cheerio highlights the problem proactively.
2. **Step 2 — Plain Language Translation:** User clicks *"Auto-Fix Rejected Template"*. Copilot explains why Meta rejected it in plain Hinglish/English: *"Missing sample value for {{1}} and promotional words inside Utility category"*.
3. **Step 3 — 1-Click AutoFix & Resubmission:** A side-by-side diff displays the exact fixes. User clicks *"Apply Auto-Fix & Resubmit to Meta"*. The system programmatically mutates the payload and pings Meta's API. Status turns green **APPROVED**.
4. **Step 4 — Edge-Case Escalation:** If the user asks about an account ban or custom ERP webhook, Copilot displays the **Warm Escalation Drawer** with pre-filled Ticket Dossier #CH-8921.

PART 3: HOW AI WAS USED & PART 4: FINANCIAL MODEL & NUMBERS

Part 3: AI Usage, Prompt Engineering & Transparency Log

Area / Prompt	Exact Prompt Strategy & Iteration	Outcome & Attribution
Prompt 1: Error Taxonomy (Worked)	*"Map the top 10 reasons non-technical Indian SMBs fail Meta Cloud API setup into a 4-part table: Meta error code, colloquial Hindi phrase, plain English cause, and 1-click programmatic fix."*	Generated the exact logic powering our 1-Click AutoFix rules engine.
Prompt 2: BPO Economics (Worked)	*"Model 5,000 SMBs comparing 12 Indian L1 agents (₹45k/mo) vs. GPT-4o-mini with prompt caching. Include formulas for Google Sheets."*	Produced our audit-ready financial model with exact cell formulas.
Prompt 3: Failed Prompt & Correction	*Initial Prompt: "Design a support chatbot for Cheerio."* (Outputted generic FAQ links). *Corrected Prompt: "Reject passive chatbots. Design an in-workflow telemetry engine that executes 1-click payload mutations."*	Shifted the product concept from a useless FAQ bot to an active diagnostics copilot.
Hallucinations Caught by Human PM	AI initially estimated 50,000 monthly tickets and frontier GPT-4 pricing ($15k/mo). Corrected to realistic SaaS volume (9,000 tickets) and cached models (₹4.2k/mo). AI also claimed it could "unban WhatsApp numbers"; corrected to official appeal packets.	Human judgment ensured technical accuracy and domain credibility.
Solely Human Judgment	User empathy for Indian SMBs facing festive deadlines; deciding basic support diagnostics must be 100% free to fuel WhatsApp messaging revenue.	Core product strategy and monetization philosophy.

PART 4: FINANCIAL MODEL — COST, BILLING, MARGIN & AUDIT (5,000 BUSINESSES)

Dimension / Parameter	Pure Human Baseline	Cheerio Copilot Hybrid	Google Sheets Formula	Net Impact / Savings
Total Monthly Inbound Queries	9,000	9,000	`=(B3*B5)+(B4*B6)`	1.8 queries / active business
Queries Resolved by AI Autonomously	0 (0%)	**5,940 (66.0%)**	`=SUM(F11:F13)`	5,940 tickets deflected
Queries Escalated to Human Agents	9,000 (100%)	3,060 (34.0%)	`=SUM(G11:G13)`	Pre-packaged dossiers attached
Average Handling Time (AHT)	18 Minutes	**7 Minutes**	Empirical benchmark	-61% handling time reduction
Full-Time Support Agents Needed	**12 Agents**	**3 Agents**	`=ROUNDUP(Queries/Cap, 0)`	**9 Support Hires Avoided**
Monthly Human Labor Cost (@ ₹45k/mo)	₹5,40,000 / mo ($6,467)	₹1,35,000 / mo ($1,617)	`=Agents*45000`	-₹4,05,000 / month
Monthly AI Running Cost (Tokens + DB)	₹0	**₹4,239 / mo ($50.77)**	`=(9000*$0.00064)+$45`	₹0.054 / query (GPT-4o-mini cached)
TOTAL MONTHLY SUPPORT EXPENDITURE	**₹5,40,000 / month**	**₹1,39,239 / month**	`=HumanCost + AICost`	**SAVE ₹4,00,761 / MO (-74.2%)**
ANNUALIZED SUPPORT EXPENDITURE	**₹64.8 Lakhs / Year**	**₹16.7 Lakhs / Year**	`=MonthlyCost * 12`	**NET ANNUAL SAVINGS: ₹48.1 LAKHS ($57.6k)**

Monetization Strategy & Audit Controls:

- **Core Diagnostic Support is 100% Free:** Cheerio monetizes on subscriptions (₹1,999+) and WhatsApp messaging volume. Free support accelerates onboarding and eliminates churn.
- **Premium Add-On ("AI Workflow Copilot"):** Prompt-to-flow campaign builder priced at **₹999/month**. With 8% adoption (400 businesses), it yields **₹3,99,600/month at a 97.5% gross margin**.
- **Audit & Quality Safeguards:** (1) Rate-limit circuit breaker at 100 queries/day/account; (2) Weekly margin reconciliation script; (3) Outage anomaly trigger if escalations exceed 40% in a 2-hr window; (4) CSAT audit prompt (👎 automatically reopens ticket).